BHARATIYA VIDYA BHAVAN'S
SARDAR PATEL INSTITUTE OF TECHNOLOGY
Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai – 400058-India
Department of Computer Engineering

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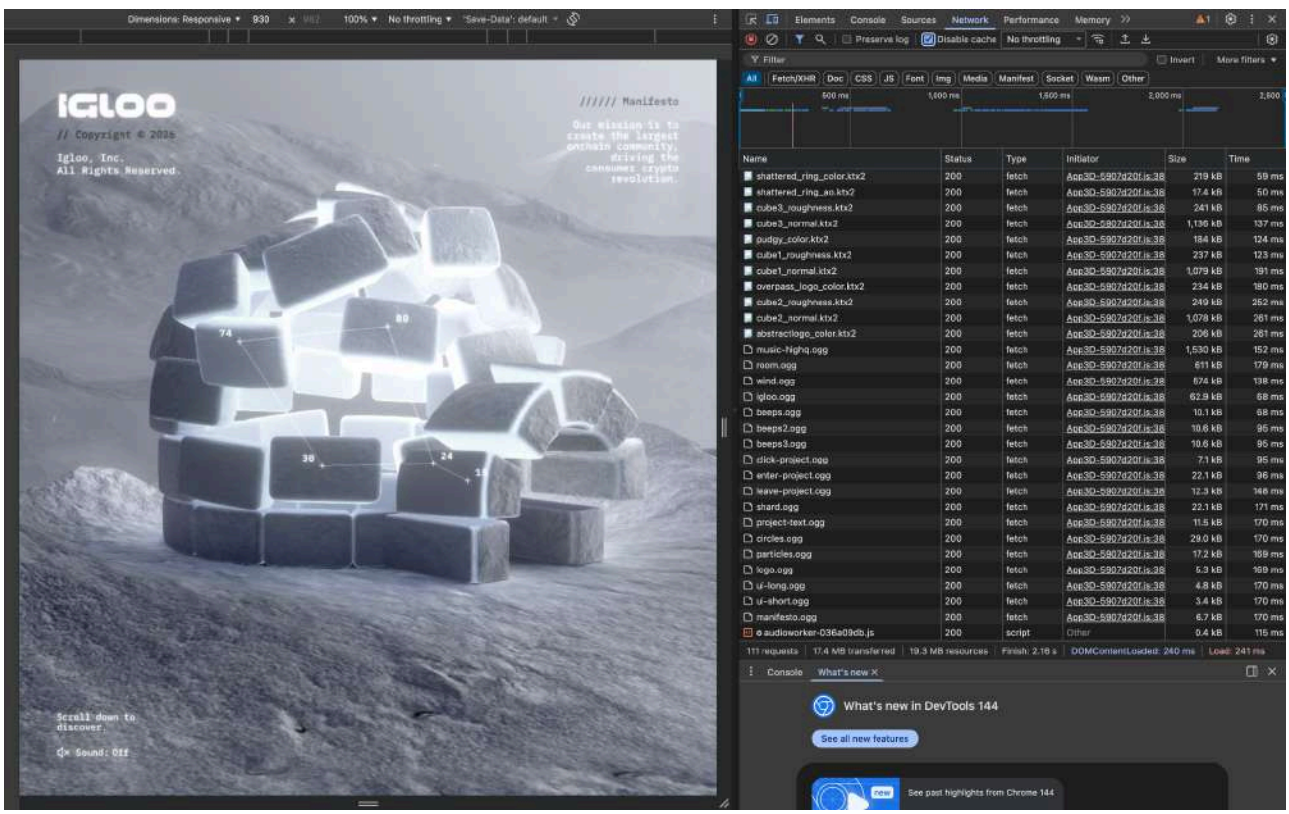
UID no.	2024300200
Experiment No.	02

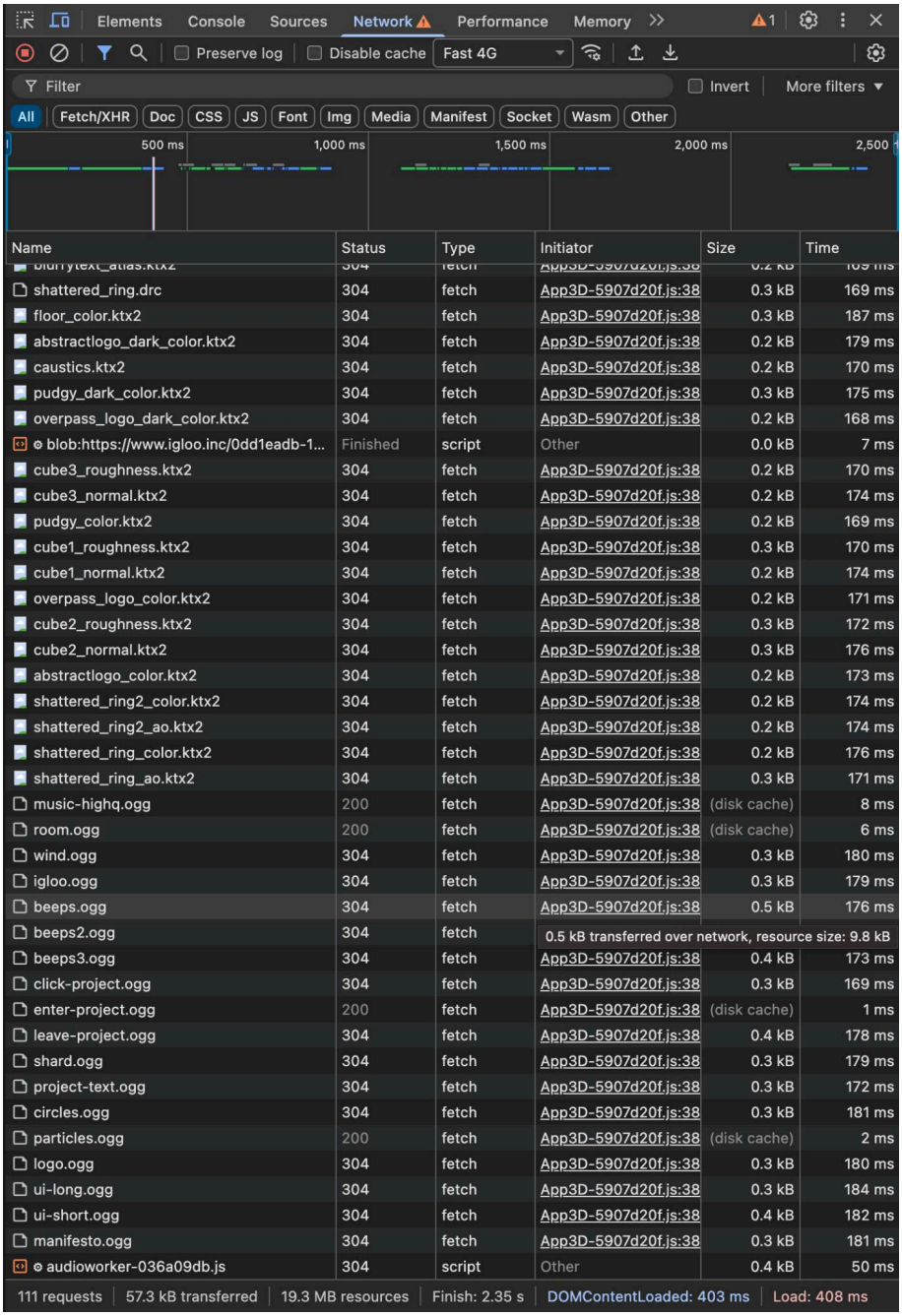
AIM:	Analyze network performance parameters using chrome devtool
TASK 1	
PROBLEM STATEMENT :	**Open Chrome DevTools**

RESULT:

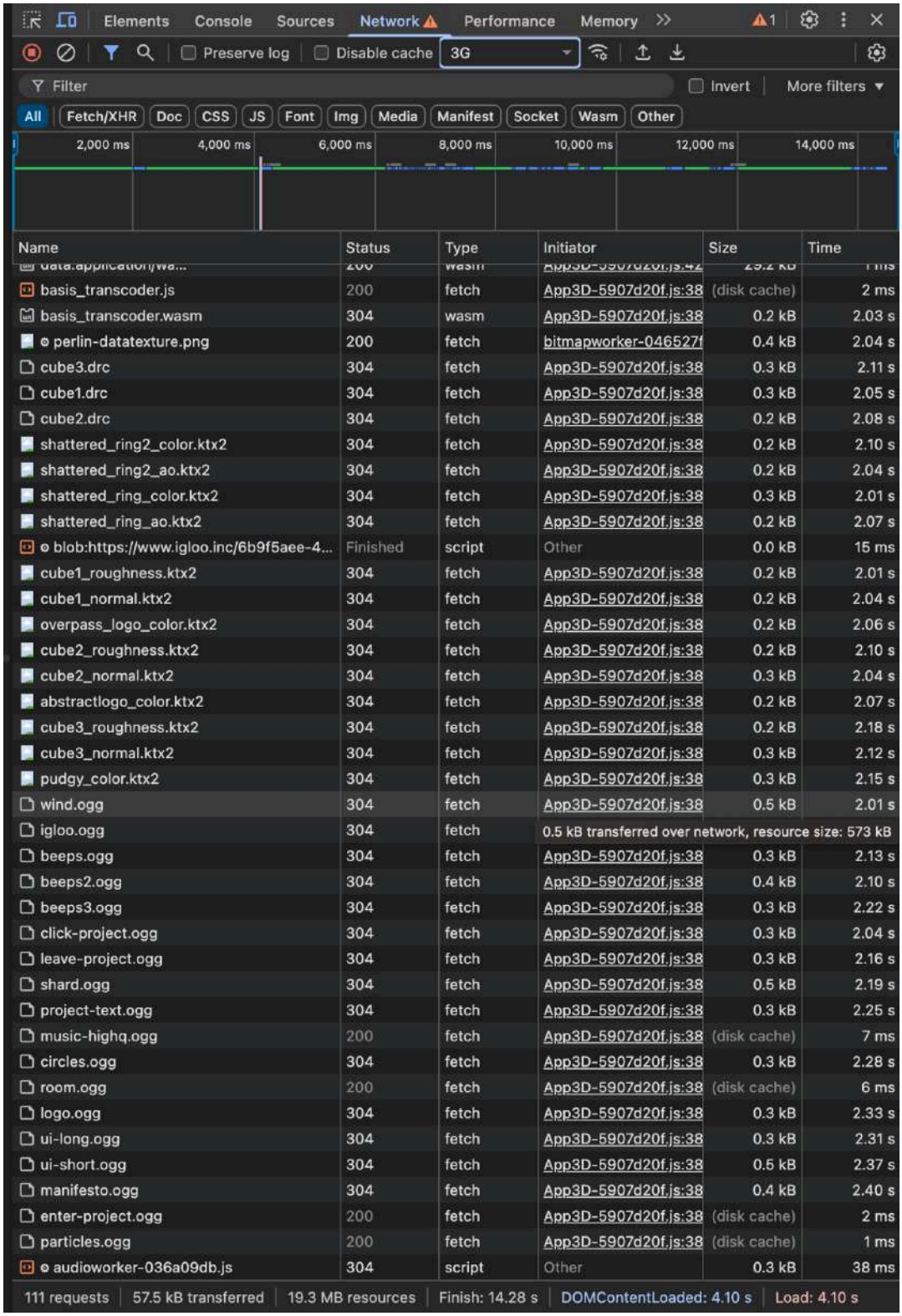


TASK 2	
PROBLEM STATEMENT :	**Clear the Network Cache**
PROGRAM:	**cmd+shit+R**

RESULT: Clearing cache removes temporarily stored data that a system, application, or network service keeps to improve speed and performance. Cache stores frequently used information so it doesn't have to be fetched again and again. Over time, this data can become outdated, corrupted, or excessive, which may cause incorrect results or performance issues.	
TASK 3 & 4	
PROBLEM STATEMENT:	Test the Page with Different Throttling Profiles
RESULT:	



Name	Status	Type	Initiator	Size	Time
shattered_ring.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	169 ms
floor_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	187 ms
abstractlogo_dark_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	179 ms
caustics.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	170 ms
pudgy_dark_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	175 ms
overpass_logo_dark_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	168 ms
blob:https://www.igloo.inc/0dd1eadb-1...	Finished	script	Other	0.0 kB	7 ms
cube3_roughness.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	170 ms
cube3_normal.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	174 ms
pudgy_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	169 ms
cube1_roughness.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	170 ms
cube1_normal.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	174 ms
overpass_logo_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	171 ms
cube2_roughness.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	172 ms
cube2_normal.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	176 ms
abstractlogo_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	173 ms
shattered_ring2_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	174 ms
shattered_ring2_ao.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	174 ms
shattered_ring_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	176 ms
shattered_ring_ao.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	171 ms
music-highq.ogg	200	fetch	App3D-5907d20f.js:38	(disk cache)	8 ms
room.ogg	200	fetch	App3D-5907d20f.js:38	(disk cache)	6 ms
wind.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	180 ms
igloo.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	179 ms
beeps.ogg	304	fetch	App3D-5907d20f.js:38	0.5 kB	176 ms
beeps2.ogg	304	fetch	0.5 kB transferred over network, resource size: 9.8 kB		
beeps3.ogg	304	fetch	App3D-5907d20f.js:38	0.4 kB	173 ms
click-project.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	169 ms
enter-project.ogg	200	fetch	App3D-5907d20f.js:38	(disk cache)	1 ms
leave-project.ogg	304	fetch	App3D-5907d20f.js:38	0.4 kB	178 ms
shard.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	179 ms
project-text.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	172 ms
circles.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	181 ms
particles.ogg	200	fetch	App3D-5907d20f.js:38	(disk cache)	2 ms
logo.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	180 ms
ui-long.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	184 ms
ui-short.ogg	304	fetch	App3D-5907d20f.js:38	0.4 kB	182 ms
manifesto.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	181 ms
audioworker-036a09db.js	304	script	Other	0.4 kB	50 ms



Name	Status	Type	Initiator	Size	Time
basis_transcoder.js	200	fetch	App3D-5907d20f.js:38	(disk cache)	2 ms
basis_transcoder.wasm	304	wasm	App3D-5907d20f.js:38	0.2 kB	2.03 s
perlin-datatexture.png	200	fetch	bitmapworker-046527f	0.4 kB	2.04 s
cube3.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.11 s
cube1.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.05 s
cube2.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.08 s
shattered_ring2_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.10 s
shattered_ring2_ao.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.04 s
shattered_ring_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.01 s
shattered_ring_ao.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.07 s
blob:https://www.igloo.inc/6b9f5aee-4...	Finished	script	Other	0.0 kB	15 ms
cube1_roughness.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.01 s
cube1_normal.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.04 s
overpass_logo_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.06 s
cube2_roughness.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.10 s
cube2_normal.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.04 s
abstractlogo_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.07 s
cube3_roughness.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	2.18 s
cube3_normal.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.12 s
pudgy_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.15 s
wind.ogg	304	fetch	App3D-5907d20f.js:38	0.5 kB	2.01 s
igloo.ogg	304	fetch	0.5 kB transferred over network, resource size: 573 kB		
beeps.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.13 s
beeps2.ogg	304	fetch	App3D-5907d20f.js:38	0.4 kB	2.10 s
beeps3.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.22 s
click-project.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.04 s
leave-project.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.16 s
shard.ogg	304	fetch	App3D-5907d20f.js:38	0.5 kB	2.19 s
project-text.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.25 s
music-highq.ogg	200	fetch	App3D-5907d20f.js:38	(disk cache)	7 ms
circles.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.28 s
room.ogg	200	fetch	App3D-5907d20f.js:38	(disk cache)	6 ms
logo.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.33 s
ui-long.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	2.31 s
ui-short.ogg	304	fetch	App3D-5907d20f.js:38	0.5 kB	2.37 s
manifesto.ogg	304	fetch	App3D-5907d20f.js:38	0.4 kB	2.40 s
enter-project.ogg	200	fetch	App3D-5907d20f.js:38	(disk cache)	2 ms
particles.ogg	200	fetch	App3D-5907d20f.js:38	(disk cache)	1 ms
audioworker-036a09db.js	304	script	Other	0.3 kB	38 ms

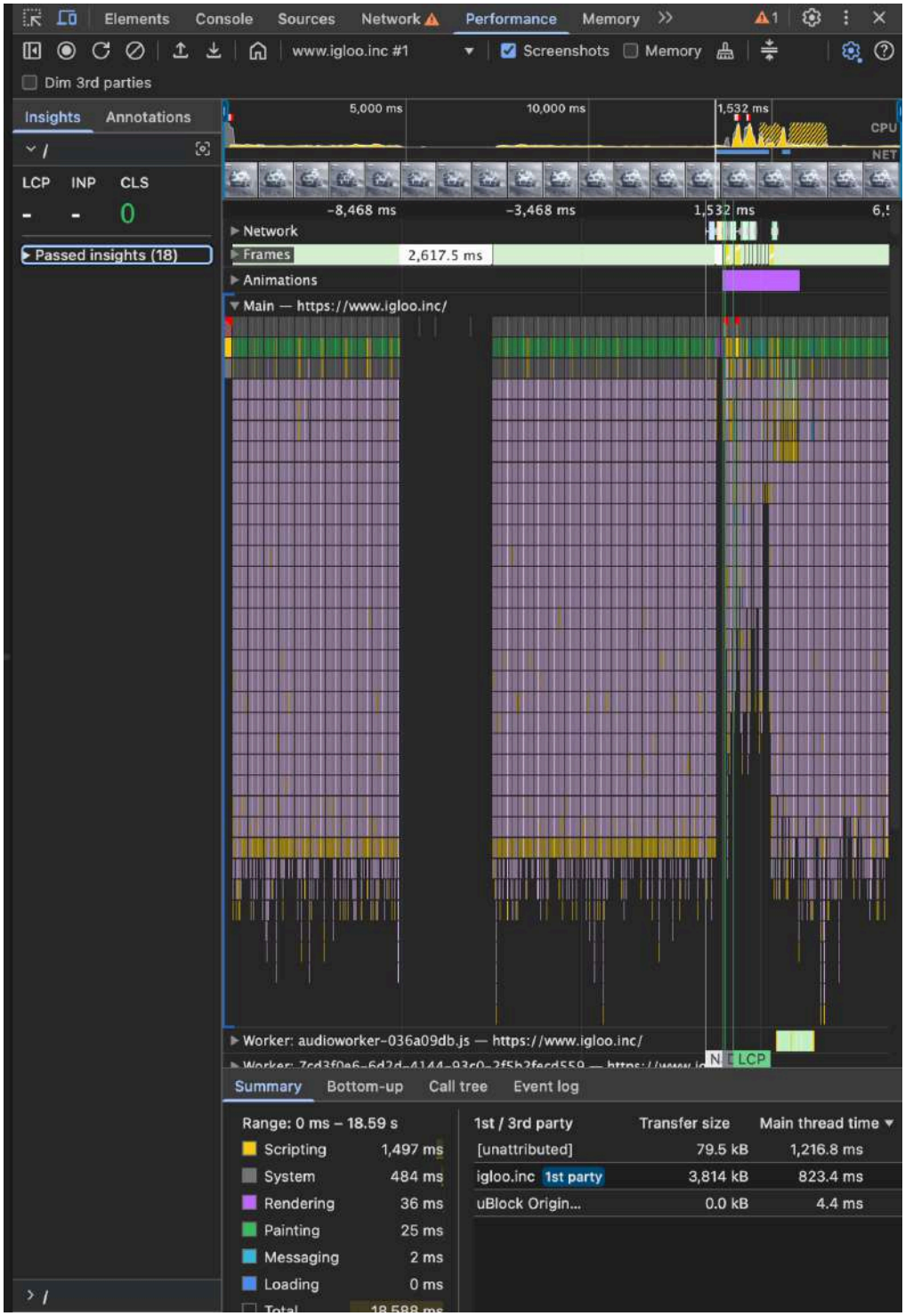
TASK 5	
PROBLEM STATEMENT:	Analyze Network Requests

RESULT:
Waterfall view:

Request/Response headers | status code | Size

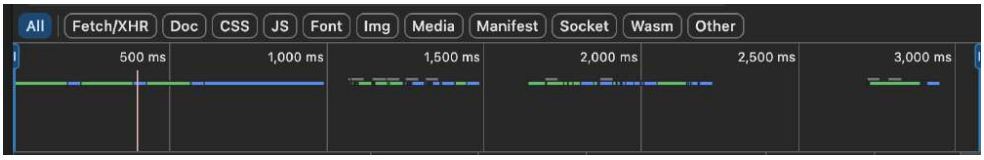
Name	Status	Type	Initiator	Size	Time
www.igloo.inc	200	document	Other	0.8 kB	173 ms
index-2d900c2d.js	200	script	(index):1	6.4 kB	179 ms
App3D-5907d20f.js	200	script	index-2d900c2d.js:141	424 kB	594 ms
favicon32-af94112f.png	200	png	Other	1.7 kB	172 ms
data:image/png;base...	200	png	App3D-5907d20f.js:41	(memory c...	0 ms
data:image/png;base...	200	png	App3D-5907d20f.js:41	(memory c...	0 ms
IBMPlexMono-Medium.json	304	fetch	App3D-5907d20f.js:38	0.3 kB	172 ms
draco_wasm_wrapper.js	200	fetch	App3D-5907d20f.js:38	(disk cache)	1 ms
draco_decoder.wasm	304	wasm	App3D-5907d20f.js:38	0.2 kB	171 ms
mountain.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	170 ms
ground.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	174 ms
igloo_cage.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	172 ms
igloo_outline.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	171 ms
igloo.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	172 ms
patch.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	174 ms
intro_particles.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	176 ms
blurrytext.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	171 ms
background_shapes.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	173 ms
shattered_ring2.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	174 ms
floor.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	173 ms
blurrytext_cylinder.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	172 ms
smoke_trail.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	173 ms
shattered_ring_smoke.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	172 ms
ceilingsmoke.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	173 ms
pudgy.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	171 ms
overpass_logo.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	174 ms
abstractlogo.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	171 ms
scroll-datatexture.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	175 ms
frost-datatexture.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	169 ms
blue-8-128-rgb.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	173 ms
logo-datatexture.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	169 ms
IBMPlexMono-Medium-datatexture.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	174 ms
sound-datatexture.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	171 ms
close-datatexture.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	175 ms
arrow-datatexture.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	174 ms
wind_noise.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	176 ms
perlin-datatexture.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	176 ms
bitmapworker-046527f8.js	304	script	Other	0.5 kB	41 ms
dot_pattern.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	177 ms

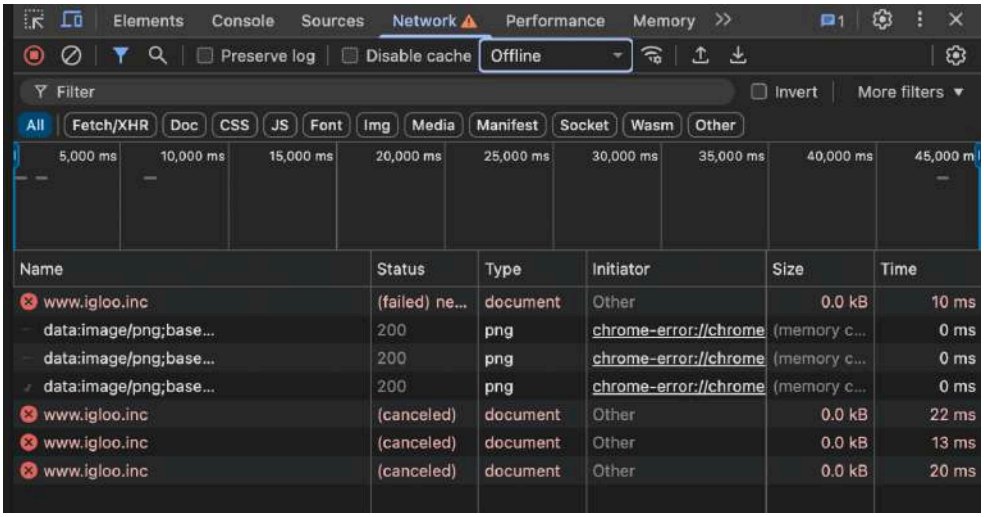
TASK 6	
PROBLEM STATEMENT:	Simulate Different Scenarios and Record Performance
RESULT:	

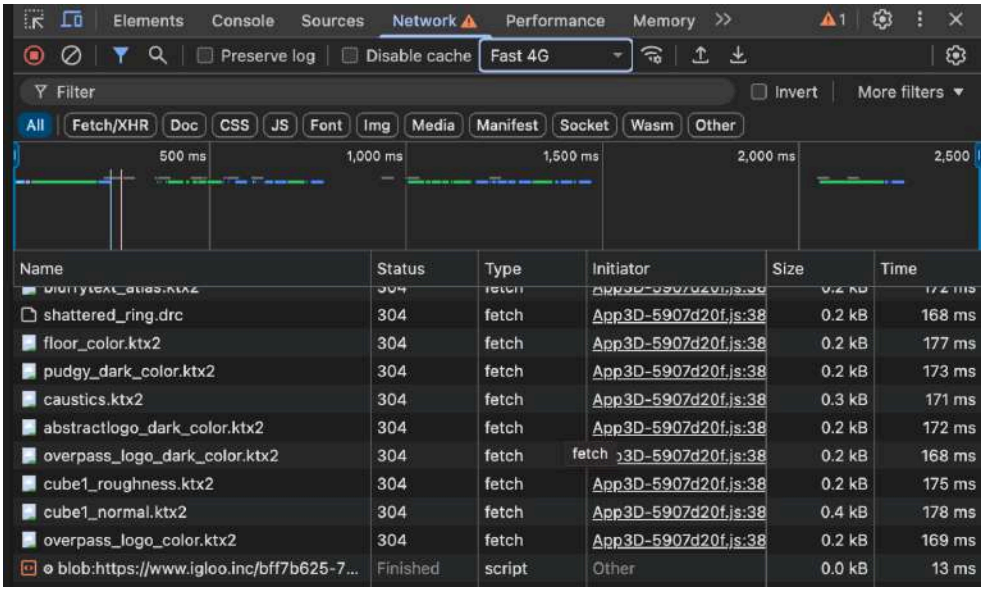

Elements
Console
Sources
Network
Performance
Memory
www.igloo.inc #1
Screenshots
Memory
Dim 3rd parties
Insights
Annotations
5,000 ms
10,000 ms
1,532 ms
CPU
NET
LCP
INP
CLS
0
Passed insights (18)
-8,468 ms
-3,468 ms
1,532 ms
Network
Frames
2,617.5 ms
Animations
Main — https://www.igloo.inc/
Worker: audioworker-036a09db.js — https://www.igloo.inc/
LCP
Summary
Bottom-up
Call tree
Event log
Range: 0 ms – 18.59 s
Scripting 1,497 ms
System 484 ms
Rendering 36 ms
Painting 25 ms
Messaging 2 ms
Loading 0 ms
1st / 3rd party
Transfer size
Main thread time
[unattributed] 79.5 kB 1,216.8 ms
igloo.inc 1st party 3,814 kB 823.4 ms
uBlock Origin... 0.0 kB 4.4 ms

QUESTIONS

Q1	**What is the purpose of network throttling in Chrome DevTools, and how does it simulate real-world conditions?**
A1	Network throttling in Chrome DevTools is used to test how a website or web application behaves under different network conditions. It helps analyze performance, loading time, and user experience when the internet connection is slow or unstable. Purpose of network throttling: 1. To evaluate website performance on slow networks 2. To observe delays in request and response transmission 3. To test user experience on mobile or low-bandwidth connections 4. To identify performance bottlenecks in web applications How it simulates real-world conditions: 5. Bandwidth limitation Download and upload speeds are reduced to simulate networks such as Slow 3G, Fast 3G, or 4G. 6. Latency introduction Artificial delay is added before sending requests and receiving responses, simulating high ping or distant servers. 7. Network congestion behavior Requests are processed more slowly, similar to real networks with limited throughput. Real-world scenarios represented: 8. Mobile data connections 9. Poor or unstable internet connectivity 10. Public Wi-Fi networks

	11. High-latency long-distance network access Network throttling in Chrome DevTools simulates real-world network conditions by limiting bandwidth and adding latency, allowing analysis of website performance and behavior under slow or unreliable network environments.
Q2	How does the "Waterfall" view in the Network tab help in understanding resource loading times?
A2	The Waterfall view in the Network tab helps in understanding how webpage resources are loaded over time and how long each resource takes to load. It provides a visual timeline that shows when a request starts, how long it takes, and how it relates to other requests. It shows the order in which different resources such as HTML, CSS, JavaScript, images, and fonts are requested by the browser. Each request is displayed as a horizontal bar whose length represents the total loading time. The Waterfall view also breaks down the loading process into different phases like DNS lookup, connection setup, request sent, waiting for the server response, and content download. By observing the Waterfall view, slow-loading or blocking resources can be easily identified. Longer bars indicate slower resources, while delays before a bar starts indicate waiting or dependency on other requests. This view also helps in understanding which resources are loaded in parallel and which ones delay the overall page load. Overall, the Waterfall view makes it easier to analyze resource loading times, detect performance bottlenecks, and understand how network conditions impact webpage performance. 

Q3	What is the difference between the "Online" and "Offline" network throttling options?
A3	The "Online" and "Offline" network throttling options in Chrome DevTools are used to control whether the browser has network connectivity while testing a website. The "Online" option means the browser is connected to the network and requests are allowed to go through normally. All resources such as HTML pages, images, scripts, and APIs can be fetched from the server based on the selected network speed or default conditions. The "Offline" option simulates a complete loss of network connectivity. When this option is enabled, the browser cannot send or receive any network requests. As a result, requests fail, and the website can only load content that is already cached or available locally. The main difference is that Online mode allows normal network communication, while Offline mode blocks all network access. Online mode is used to test performance under real network conditions, whereas Offline mode is used to test how a website behaves when there is no internet connection, such as error handling, offline pages, or cached content loading. 

<table>
<tr><td></td><td> 
</td></tr>
<tr><td>Q4</td><td>Explain what happens when you choose different network profiles like "Fast 3G", "Regular 3G" and "Slow 3G" How do these profiles affect page load times?</td></tr>
<tr><td>A4</td><td>When different network profiles like Fast 3G, Regular 3G, and Slow 3G are selected in Chrome DevTools, the browser simulates varying mobile network speeds and delays.

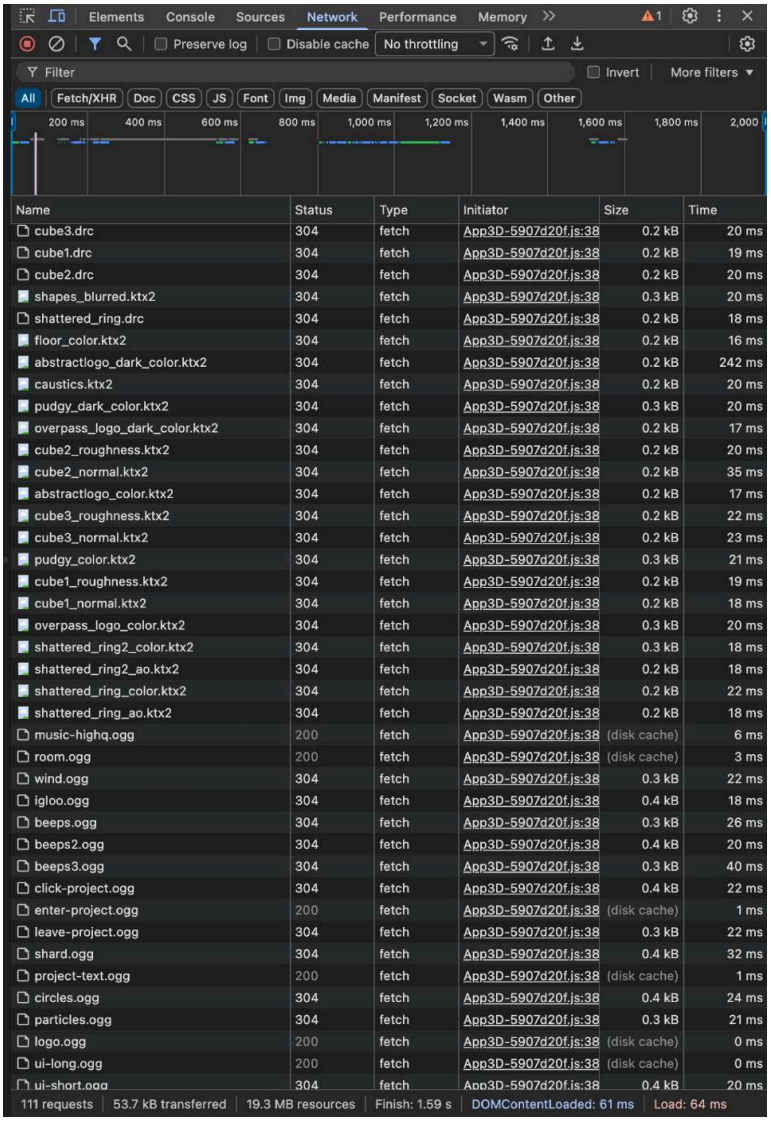
Fast 3G provides higher bandwidth and lower latency, so pages load relatively quickly. Regular 3G has moderate speed and latency, resulting in noticeable delays. Slow 3G has very low bandwidth and high latency, causing significantly longer page load times.

Overall, slower network profiles increase delays and resource loading time, leading to slower page rendering.

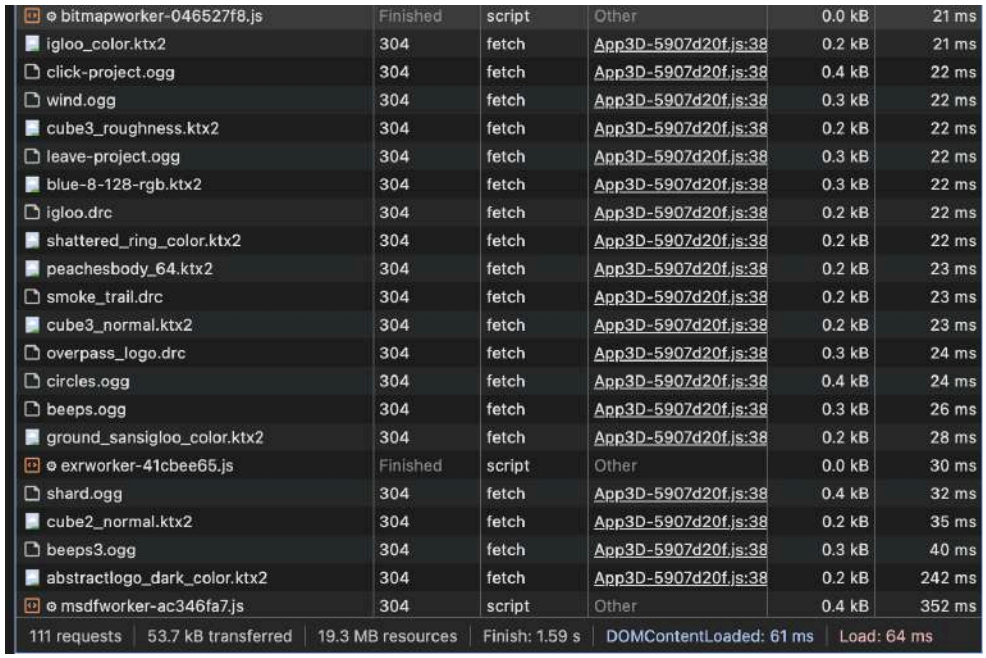
FAST 4G

3G
</td></tr>
<tr><td>Q5</td><td>What is the average load time of the webpage without any network</td></tr>
</table>

	throttling?
A5	This value depends on the specific webpage, device, and internet connection, so there is no single fixed average load time. Without any network throttling, the average load time is the actual time taken by the page to load under normal network conditions. It can be observed in Chrome DevTools by opening the Network tab, keeping throttling set to "Online", reloading the page, and noting the total load time shown at the bottom or in the Waterfall timeline. In general, for a normal webpage on a stable broadband connection, the load time is typically a few seconds, but the exact value should be recorded based on your observation during the experiment.

<table>
<tr><td></td><td>


</td></tr>
<tr><td>Q6</td><td>After applying the "Slow 3G" network profile, how does the page load time change compared to the normal network?</td></tr>
<tr><td>A6</td><td>After applying the Slow 3G network profile, the page load time increases significantly compared to the normal network due to reduced bandwidth and higher latency.
</td></tr>
<tr><td>Q7</td><td>Which resources (e.g., images, scripts, CSS files) take the longest to load when the page is throttled with "Regular 3G"?</td></tr>
</table>

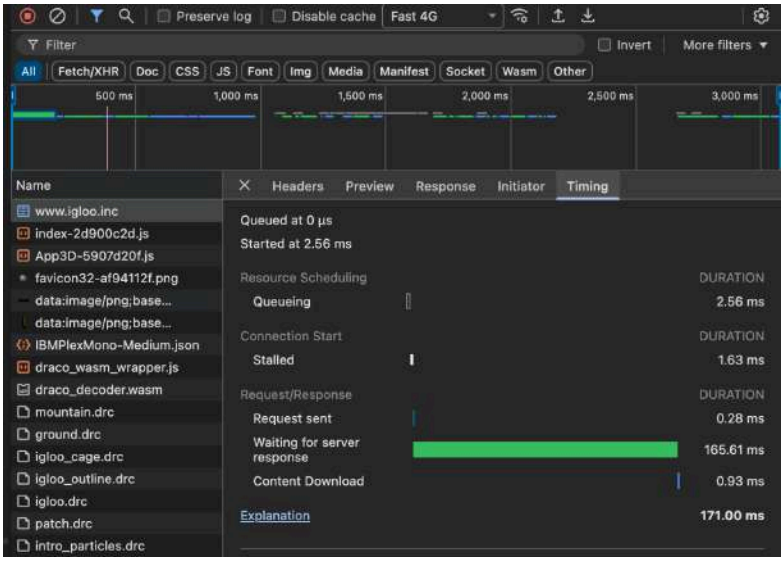
A7	
Q8	Identify any 404 or other errors in the network log when loading the page with "Slow 3G" applied. What might cause these errors?
A8	When loading the page with the Slow 3G network profile applied, 404 or other errors may appear in the network log. A 404 error indicates that a requested resource was not found on the server, which can be caused by incorrect URLs, missing files, or server-side issues. Other errors, such as failed or timed-out requests, may occur due to high latency, very low bandwidth, or request timeouts under slow network conditions.
Q9	Does the page load faster or slower with "Fast 3G" compared to "Slow

bitmapworker-046527f8.js	Finished	script	Other	0.0 kB	21 ms
igloo_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	21 ms
click-project.ogg	304	fetch	App3D-5907d20f.js:38	0.4 kB	22 ms
wind.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	22 ms
cube3_roughness.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	22 ms
leave-project.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	22 ms
blue-8-128-rgb.ktx2	304	fetch	App3D-5907d20f.js:38	0.3 kB	22 ms
igloo.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	22 ms
shattered_ring_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	22 ms
peachesbody_64.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	23 ms
smoke_trail.drc	304	fetch	App3D-5907d20f.js:38	0.2 kB	23 ms
cube3_normal.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	23 ms
overpass_logo.drc	304	fetch	App3D-5907d20f.js:38	0.3 kB	24 ms
circles.ogg	304	fetch	App3D-5907d20f.js:38	0.4 kB	24 ms
beeps.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	26 ms
ground_sansigloo_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	28 ms
exrworker-41cbee65.js	Finished	script	Other	0.0 kB	30 ms
shard.ogg	304	fetch	App3D-5907d20f.js:38	0.4 kB	32 ms
cube2_normal.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	35 ms
beeps3.ogg	304	fetch	App3D-5907d20f.js:38	0.3 kB	40 ms
abstractlogo_dark_color.ktx2	304	fetch	App3D-5907d20f.js:38	0.2 kB	242 ms
msdfworker-ac346fa7.js	304	script	Other	0.4 kB	352 ms

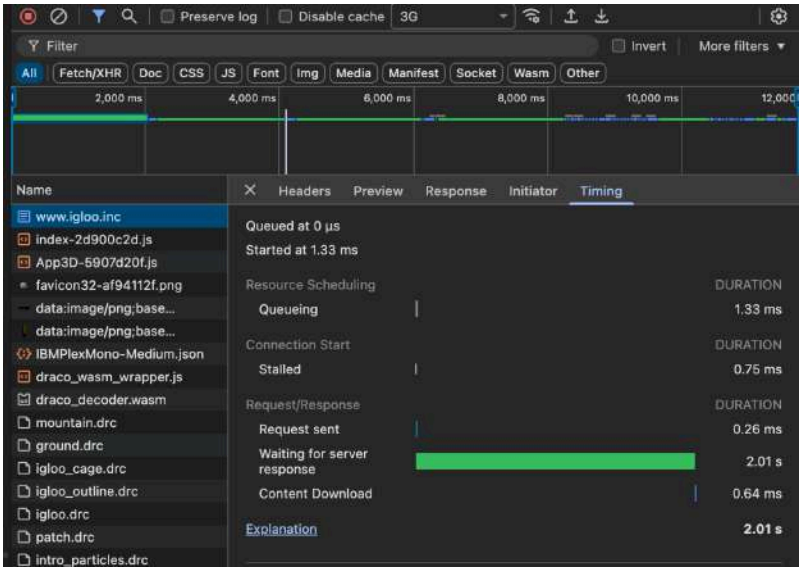
111 requests | 53.7 kB transferred | 19.3 MB resources | Finish: 1.59 s | DOMContentLoaded: 61 ms | Load: 64 ms

Name	Status
basis_transcoder.js	200
basis_transcoder.wasm	304
perlin-datatexture.png	200
cube3.drc	304
cube1.drc	304
cube2.drc	304
shattered_ring2_color.ktx2	304
shattered_ring2_ao.ktx2	304
shattered_ring_color.ktx2	304
shattered_ring_ao.ktx2	304
blob:https://www.igloo.inc/6b9f5aee-4...	Finished
cube1_roughness.ktx2	304
cube1_normal.ktx2	304

	3G"? What are the reasons for the differences?
A9	The page loads faster with Fast 3G compared to Slow 3G. This is because Fast 3G provides higher bandwidth and lower latency, allowing resources to be downloaded more quickly and requests to be processed with less delay, whereas Slow 3G has limited bandwidth and high latency, which increases loading time.
Q10	What is the "Time to First Byte (TTFB)" for the webpage under different throttling profiles? How does network throttling affect TTFB?
A10	Time to First Byte (TTFB) is the time taken from sending a request to receiving the first byte of the response from the server. Under different throttling profiles, TTFB changes as follows. With no throttling or a normal network, TTFB is the lowest because bandwidth is high and latency is low. With Fast 3G, TTFB increases slightly due to added latency. With Slow 3G, TTFB increases significantly because of high latency and delayed request–response transmission. Network throttling affects TTFB mainly by increasing latency. Slower network profiles introduce delays before the request reaches the server and before the response begins, resulting in higher TTFB values. FAST 4G

<table>
<tr><td></td><td>


3G



</td></tr>
<tr><td>Q11</td><td>How does the "DOMContentLoaded" time change when testing under different throttling profiles?</td></tr>
<tr><td>A11</td><td>Under throttling, the DOMContentLoaded time increases. Slower network speeds delay resource downloads, slowing DOM parsing and event firing.</td></tr>
</table>

	CPU throttling slows JavaScript execution, also delaying DOM construction. The more severe the throttling, the higher the DOMContentLoaded time. Fast 4G DOMContentLoaded: 403 ms 3G DOMContentLoaded: 4.09 s
Q12	What is the impact of throttling on the "Load" event timing (when the page is fully loaded)?
A12	Throttling increases the Load event timing. Slower network speeds delay downloading all resources like images, scripts, and styles, while CPU throttling slows processing and rendering. The more severe the throttling, the longer it takes for the Load event to fire.
Q13	Is there any noticeable delay in rendering when the page is loaded under "Slow 3G" conditions? If so, what resources are responsible for the delay?
A13	Yes, there is a noticeable delay under Slow 3G. Large resources like images, videos, and scripts take longer to download, which delays DOM construction and rendering. External CSS and fonts can also block rendering, contributing to the slowdown. These are the responsible resources:

<table>
<tr><td></td><td>
<table>
<tr><td>bitmapworker-046527f8.js</td><td>Finished</td><td>script</td><td>Other</td><td>0.0 kB</td><td>21 ms</td></tr>
<tr><td>igloo_color.ktx2</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.2 kB</td><td>21 ms</td></tr>
<tr><td>click-project.ogg</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.4 kB</td><td>22 ms</td></tr>
<tr><td>wind.ogg</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.3 kB</td><td>22 ms</td></tr>
<tr><td>cube3_roughness.ktx2</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.2 kB</td><td>22 ms</td></tr>
<tr><td>leave-project.ogg</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.3 kB</td><td>22 ms</td></tr>
<tr><td>blue-8-128-rgb.ktx2</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.3 kB</td><td>22 ms</td></tr>
<tr><td>igloo.drc</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.2 kB</td><td>22 ms</td></tr>
<tr><td>shattered_ring_color.ktx2</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.2 kB</td><td>22 ms</td></tr>
<tr><td>peachesbody_64.ktx2</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.2 kB</td><td>23 ms</td></tr>
<tr><td>smoke_trail.drc</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.2 kB</td><td>23 ms</td></tr>
<tr><td>cube3_normal.ktx2</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.2 kB</td><td>23 ms</td></tr>
<tr><td>overpass_logo.drc</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.3 kB</td><td>24 ms</td></tr>
<tr><td>circles.ogg</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.4 kB</td><td>24 ms</td></tr>
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<tr><td>ground_sansigloo_color.ktx2</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.2 kB</td><td>28 ms</td></tr>
<tr><td>exrworker-41cbee65.js</td><td>Finished</td><td>script</td><td>Other</td><td>0.0 kB</td><td>30 ms</td></tr>
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<tr><td>abstractlogo_dark_color.ktx2</td><td>304</td><td>fetch</td><td>App3D-5907d20f.js:38</td><td>0.2 kB</td><td>242 ms</td></tr>
<tr><td>msdfworker-ac346fa7.js</td><td>304</td><td>script</td><td>Other</td><td>0.4 kB</td><td>352 ms</td></tr>
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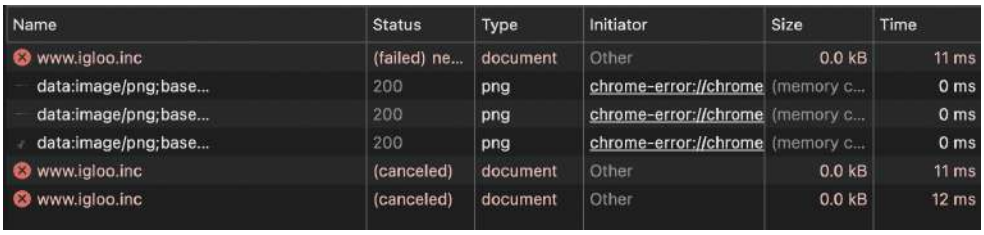
</table>
</td></tr>
<tr><td>Q14</td><td>Are there any large resources (e.g., images, scripts) that cause the page to load slowly under network throttling? Identify them.</td></tr>
<tr><td>A14</td><td>
<table>
<tr><td>bitmapworker-046527f8.js</td><td>Finished</td><td>script</td><td>Other</td><td>0.0 kB</td><td>21 ms</td></tr>
<tr><td>exrworker-41cbee65.js</td><td>Finished</td><td>script</td><td>Other</td><td>0.0 kB</td><td>30 ms</td></tr>
<tr><td>msdfworker-ac346fa7.js</td><td>304</td><td>script</td><td>Other</td><td>0.4 kB</td><td>352 ms</td></tr>
</table>
</td></tr>
<tr><td>Q15</td><td>What effect does using the "Cache" have on performance when throttling is applied? Does the page load faster on subsequent visits?</td></tr>
<tr><td>A15</td><td>Using the cache improves performance even under throttling. Cached resources don't need to be re-downloaded, so the browser can render the page faster. Yes, the page typically loads significantly faster on subsequent visits because fewer network requests are needed.</td></tr>
<tr><td>Q16</td><td>What happens to page load time when you simulate no connection ("Offline")? Which resources fail to load?</td></tr>
<tr><td>A16</td><td>When simulating offline, page load time increases dramatically or the page may fail to load completely. All resources that require a network—like</td></tr>
</table>

	images, scripts, stylesheets, fonts, and API requests—fail to load unless they are cached or available via service workers.  Name | Status | Type | Initiator | Size | Time www.igloo.inc | (failed) ne... | document | Other | 0.0 kB | 11 ms data:image/png;base... | 200 | png | chrome-error://chrome | (memory c... | 0 ms data:image/png;base... | 200 | png | chrome-error://chrome | (memory c... | 0 ms data:image/png;base... | 200 | png | chrome-error://chrome | (memory c... | 0 ms www.igloo.inc | (canceled) | document | Other | 0.0 kB | 11 ms www.igloo.inc | (canceled) | document | Other | 0.0 kB | 12 ms
Q17	How does switching to a custom throttling profile with a higher latency (e.g., 200ms) affect the loading of the page compared to "Regular 3G"?
A17	Switching to a custom throttling profile with higher latency increases page load times compared to Regular 3G. Each request takes longer to reach the server and receive a response, delaying the download of resources, DOM construction, and overall rendering. The higher the latency, the slower the page feels to the user.
CONCLUSION:	In this experiment, we explored how different network conditions affect webpage performance using Chrome DevTools. By applying network throttling profiles like Fast 3G, Regular 3G, and Slow 3G, we observed changes in page load times, resource loading sequences, and Time to First Byte (TTFB). The Waterfall view helped identify slow or blocking resources, while request and response headers provided insights into server communication. Overall, the experiment demonstrated that slower network speeds and higher latency significantly increase load times and affect user experience, highlighting the importance of optimizing websites for varying real-world network conditions.